

Name: _____

Class: _____

Date: _____

Lab 11: List Algorithms

Python Lab · Section 3.9–3.10 Algorithms & Lists



Goal

Write the classic list algorithms by hand: sum, average, max, min, linear search, and filter.

Warm-up: Predict

```
nums = [4, 9, 2]
biggest = nums[0]
for n in nums:
    if n > biggest:
        biggest = n
print(biggest)
```

Predict the output. Why start biggest at `nums[0]`?

AP Connection

- These exact algorithms appear on the AP exam.
- Start max/min at the first element, not 0!

Build It

Step 1. Sum and average without `sum()`:

```
scores = [88, 92, 75, 64, 99]
total = 0
for s in scores:
    total = total + s
print(total / len(scores))
```

Step 2. Max and min without `max()/min()`. Test with a list of all negative numbers!

```
smallest = scores[0]
for s in scores:
    if s < smallest:
        smallest = s
```

Step 3. Linear search: does the list contain a target?

```
target = int(input("Find: "))
found = False
for s in scores:
    if s == target:
        found = True
print(found)
```

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Build It (continued) · Level Up

Build It (continued)

Step 4. Filter: build a new list of passing scores (≥ 70).

```
passing = []
for s in scores:
    if s >= 70:
        passing.append(s)
print(passing)
```

Level Up Challenges

Mild Count how many scores are above the average.

Medium Return the INDEX where the target is found (or -1).

Spicy Remove every score below 70 from the original list while looping. Watch out for skipped elements!

Python → AP Pseudocode

```
passing = []
for s in scores:
    if s >= 70:
        passing.append(s)
```

Write it in AP pseudocode.

Exit Ticket

Why does starting max at 0 fail for some lists?
